

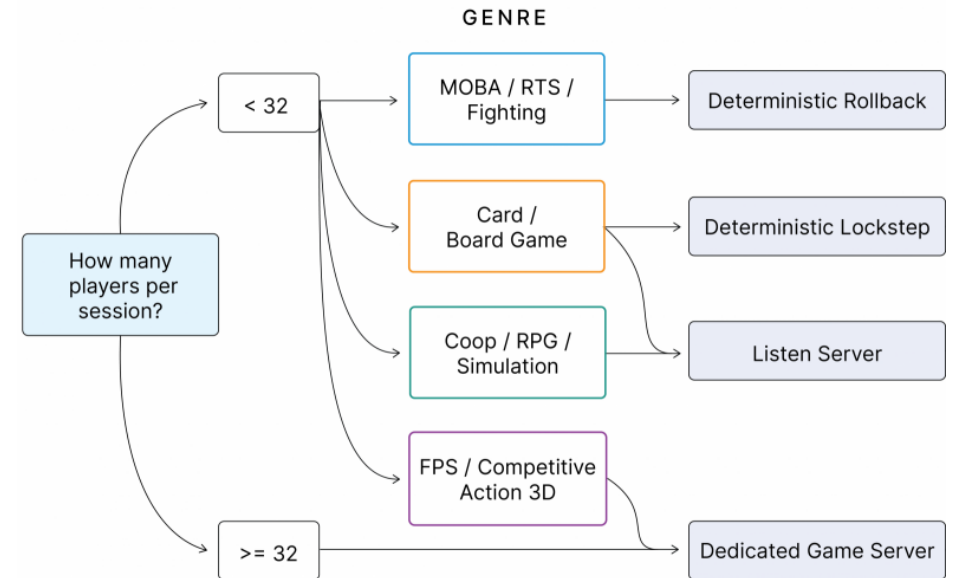
Video Game Programming: Multiplayer Game Programming

L19

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Multiplayer games

- Multiplayer games allow players to cooperate or compete
- Several genres (from FPS to card games) can benefit from multiplayer
- All online multiplayer games should deal with network-related challenges, such as latency and packet loss
- Suitable network implementation and API depends on game's genre, number of players,...



Network implementations

- **Dedicated game server** is a client-server network implementation where the server is hosted on dedicated computers; expensive, but scalable and secure
- **Listen server** is a client-server implementation where the server is hosted on a client device; inexpensive, but not scalable and not secure
- **Deterministic lockstep** is a P2P implementation where only inputs are sent to all other players and synchronized in a “locked step” and determinism on each client ensures that they all stay at the same state; inexpensive and secure, but with complex determinism
- **Deterministic rollback** is an enhancement of the previous where clients forward-predict inputs while waiting for updates; relatively inexpensive and secure, but with very complex determinism and simulation

Multiplayer APIs for Unity

- Unity had a multiplayer API (UNet), now deprecated
- A new mid-level API (MLAPI, now called Netcode) is developed as an open-source project (<https://docs-multiplayer.unity3d.com/>)
- The old high-level API was available up to 2023.1, but Unity recommended MLAPI or third-party solutions since 2018.3 (see blog post in useful resources)
- A shared idea between solutions is the use of a network manager to handle network players spawning
- Multiplayer games also usually need a service to match players

Types of Multiplayer games based on topology

Local Multiplayer

Cuphead
Mario Kart
Tricky Towers
Wii Sport
Untitled Goose Game
It takes two



Peer to Peer (P2P)

Elden Ring
GTA Online
F1 2021
Super Smash Bros
Hell Divers 2
Risk of Rain 2



Dedicated Server

Fortnite
League of Legends
World of Warcraft
Final Fantasy XIV
Guild Wars 2
Minecraft



Local Multiplayer

Pros

- No latency
- No costs
- No need to connect to the internet
- Simplest type

Cons

- Low scalability
- All players must be in the same room
- Split screen



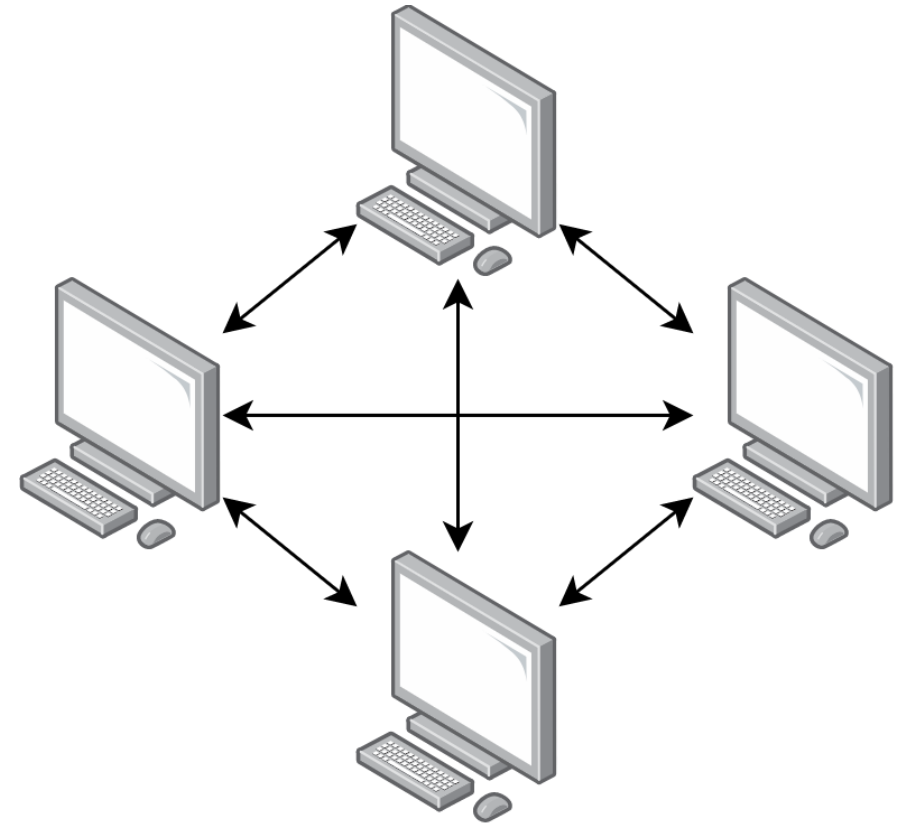
Peer To Peer (Direct P2P)

Pros

- Connect users across the Internet
- Low cost (possible costs for connecting players)
- Lowest latency

Cons

- Complexity of development
- Low Scalability
- Security (all clients know all IPs)
- Not permitted on some platforms for policy reasons



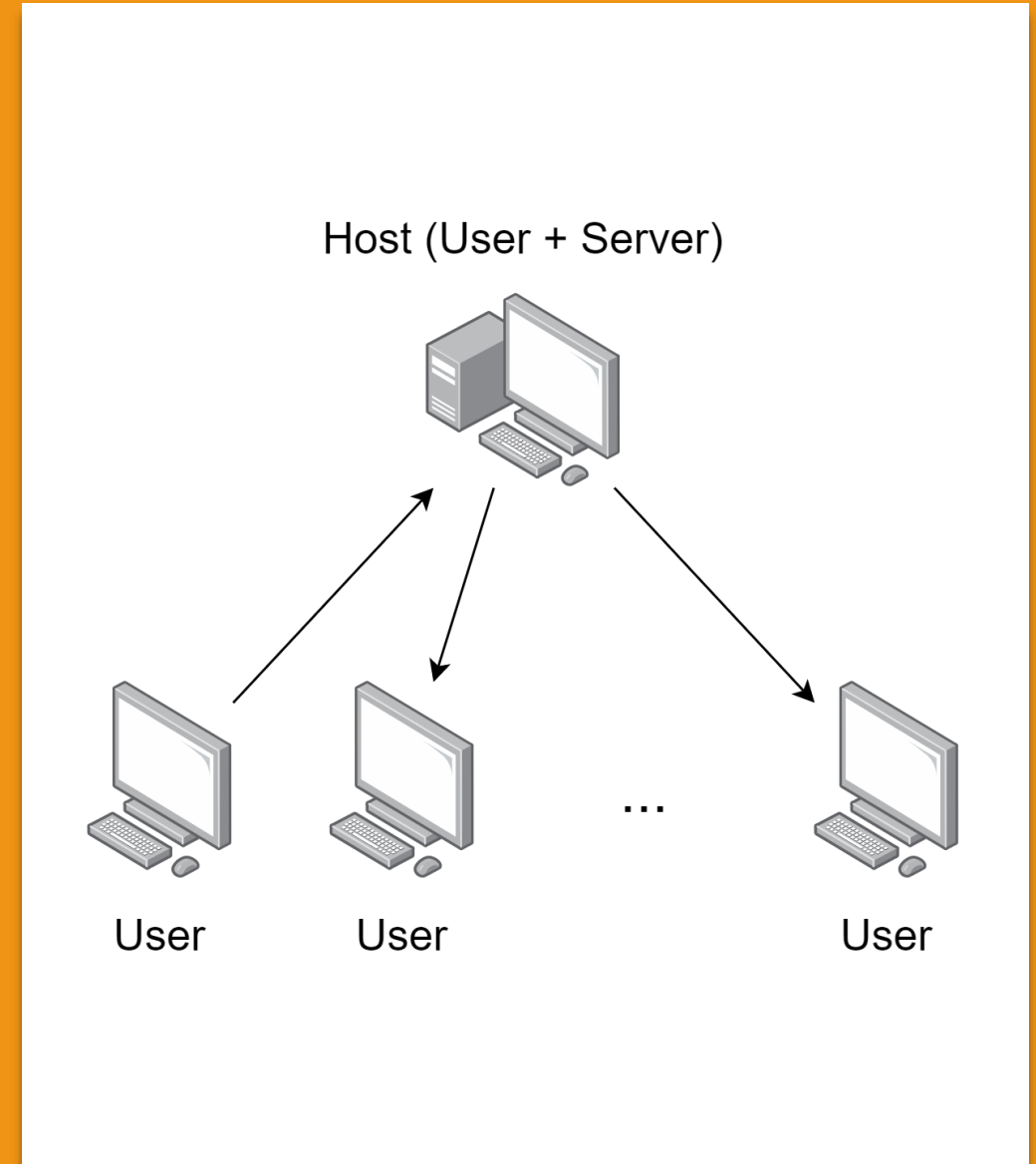
Peer To Peer (Player-Hosted client / server)

Pros

- Connect users across the Internet
- Low cost (possible costs for connecting players)
- More secure than Direct P2P

Cons

- Host is in advantage (no latency)
- Host migration (if drops connection)
- Host IP needs to be sent to other clients
- Host can control the server (cheating)
- Low Scalability (based on host bandwidth and computer power)



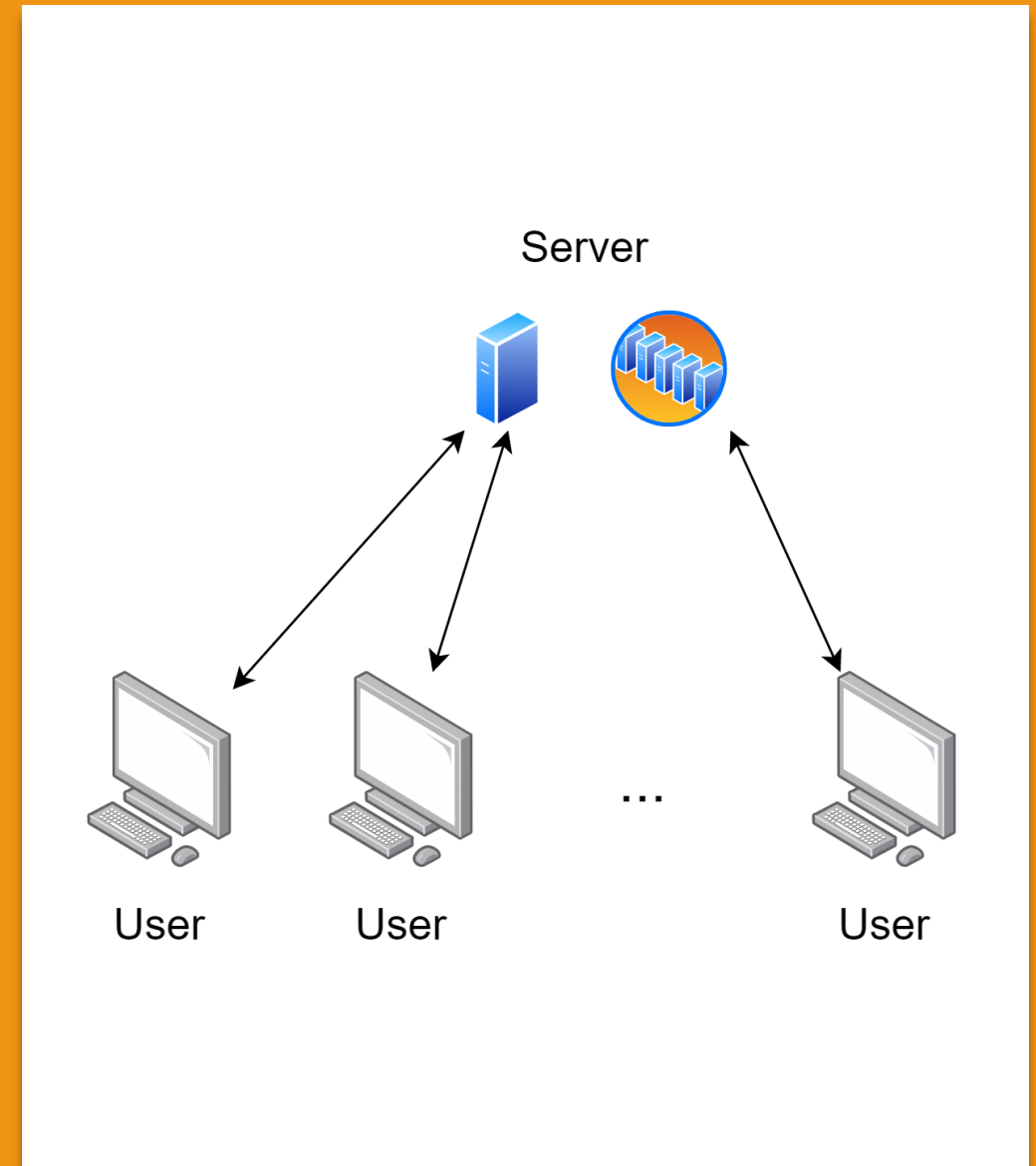
Dedicated Server

Pros

- Scalability
- Performance (can scale up to the number of players)
- Global distribution (can locate a server in a nearest region to lower latency)
- Security (only servers IPs are known to players)
- No host advantage
- Reliability

Cons

- Complexity of development
- Cost for bandwidth and servers



Unity Game Services (UGS)

A complete service ecosystem for live games



Build your foundation

Accounts

Enable across-platform player sign-in and authentication system.

Multiplayer

Get scalable and performant multiplayer game hosting.

Content management

Deploy and manage your game content.

DevOps

Explore version control and build automation built for game development.



Engage your players

Analytics

Drive data-based decision making with analytics tools.

Player engagement

Optimize the player experience with experimentation and engagement tools.

Community

Connect players and keep them safe with scalable voice and text chat.

Crash reporting

Improve game stability with crash reporting and investigation tools.



Grow your game

Monetization

Drive revenue with in-game ads.

User acquisition

Find the right users to grow your audience.

Ad mediation

Increase ad demand and generate more revenue from your game.

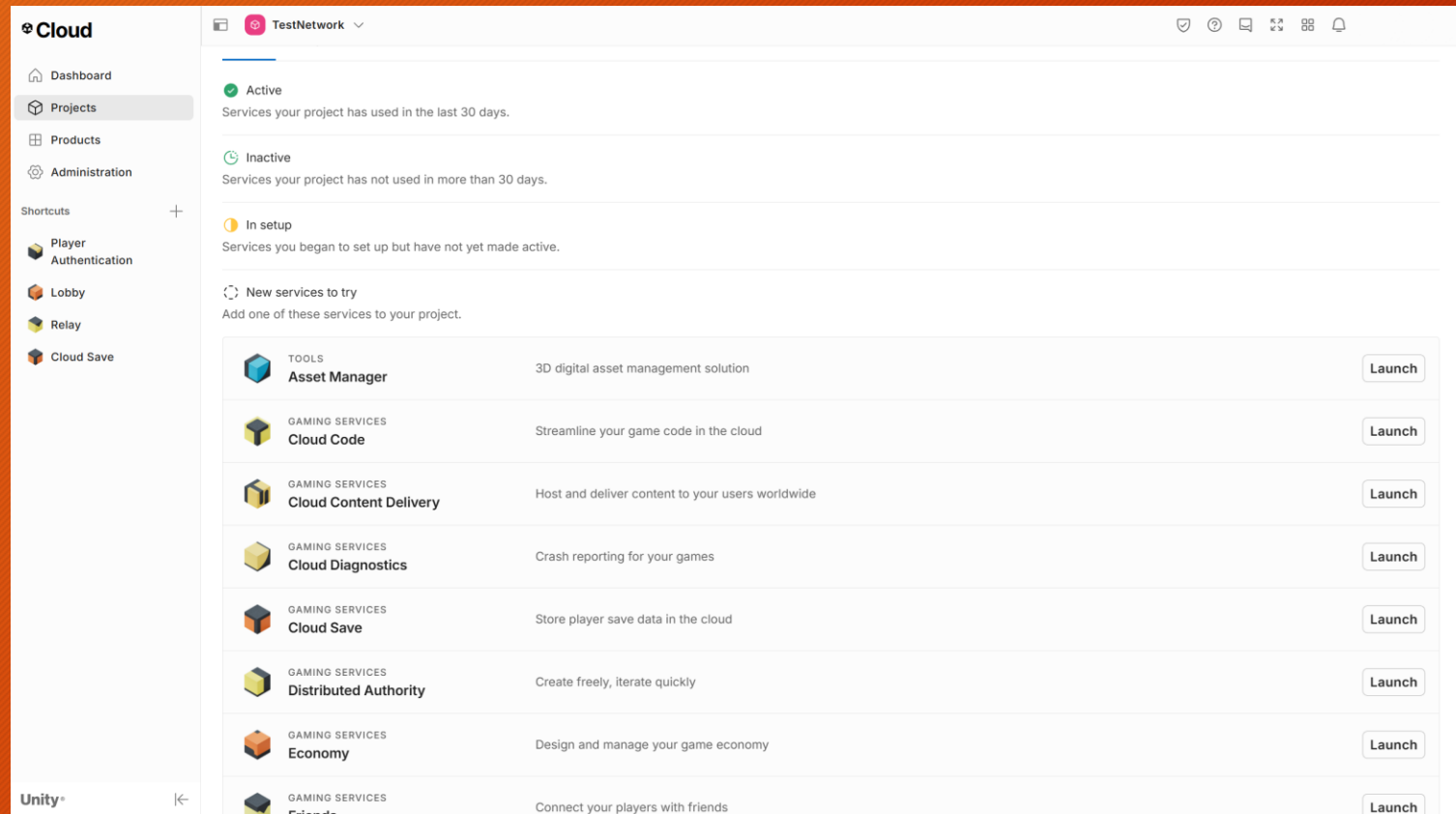
Publishing

Turn your mobile game idea into a business.

Game economy

Design your in-game economy infrastructure and add in-app purchases.

Unity Cloud Services Project View

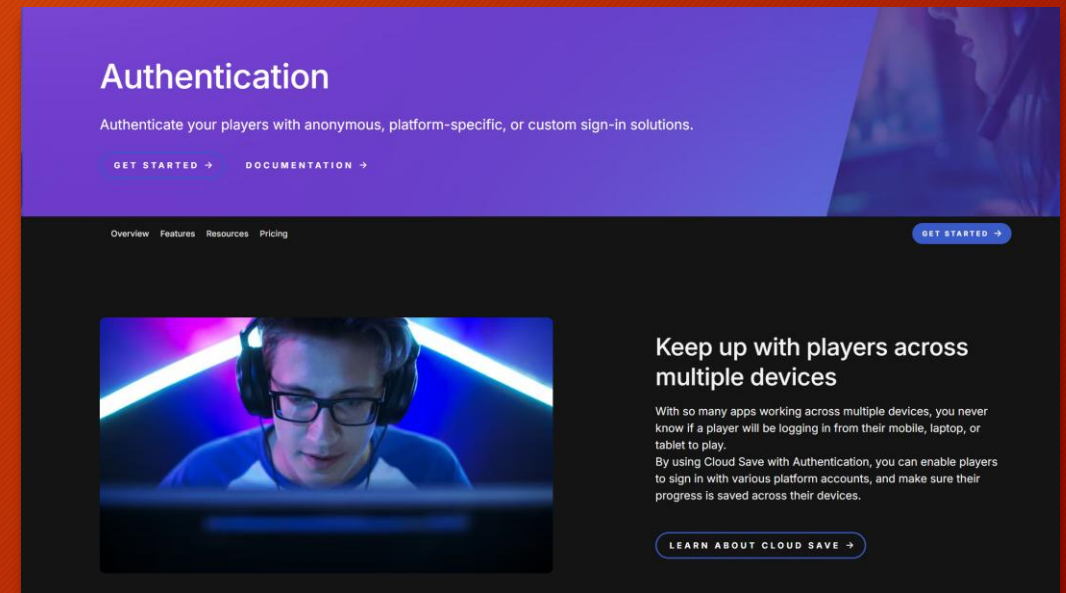


Remote Multiplayer Requirements

- Authentication of the players
- Matchmaking
- Connect players with low latency
- Persistent data (save player's progression in the game, items, skins, ...)
- Analytics (active users, returning users, monetization, ...)

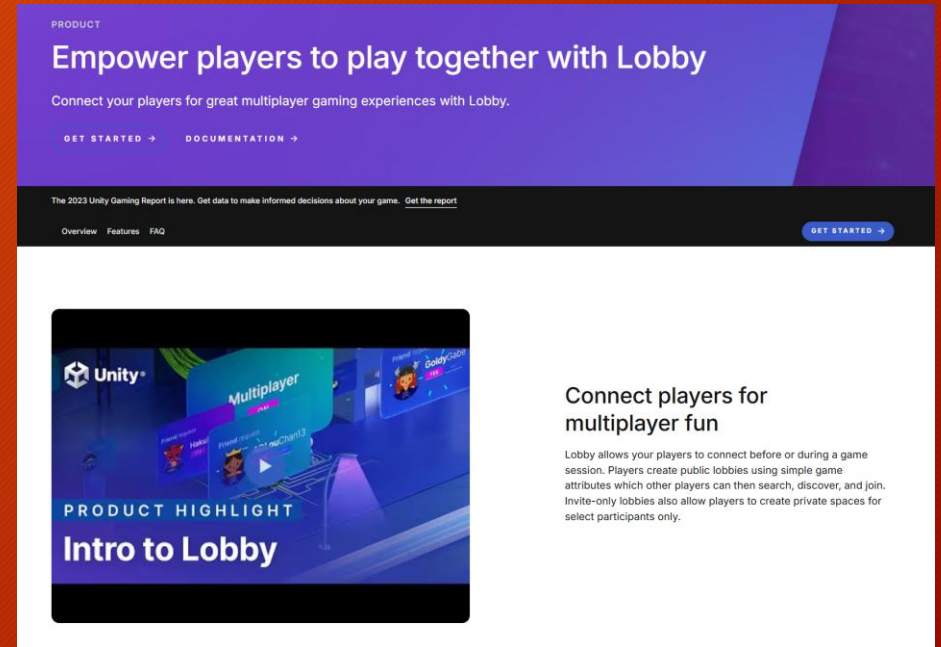
Authentication with Player Authentication

- <https://unity.com/products/authentication>
- Needed to authenticate players and use other UGS
- Can be anonymous
- Can be platform-specific, for example:
 - Username and password
 - Steam
 - Google
 - Google Play Games
 - Apple
 - Apple Game Center
 - Oculus



Matchmaking with Lobby

- <https://unity.com/products/lobby>
- Unity provides global matchmaking with Lobby
- Private and public lobbies
- Quick join lobby



The screenshot shows the Unity Lobby product page. The header is purple with the text 'PRODUCT Empower players to play together with Lobby' and a sub-header 'Connect your players for great multiplayer gaming experiences with Lobby.' Below this are links for 'GET STARTED' and 'DOCUMENTATION'. A dark blue bar contains the text 'The 2023 Unity Gaming Report is here. Get data to make informed decisions about your game. Get the report' and a 'GET STARTED' button. The main content area features a video player with the Unity logo, 'Multiplayer' text, and 'PRODUCT HIGHLIGHT Intro to Lobby'. To the right of the video, the text 'Connect players for multiplayer fun' is followed by a paragraph explaining that Lobby allows players to connect before or during a game session, create public lobbies, and search for other players.

PRODUCT

Empower players to play together with Lobby

Connect your players for great multiplayer gaming experiences with Lobby.

[GET STARTED](#) → [DOCUMENTATION](#) →

The 2023 Unity Gaming Report is here. Get data to make informed decisions about your game. [Get the report](#)

[Overview](#) [Features](#) [FAQ](#) [GET STARTED](#) →

Unity

Multiplayer

PRODUCT HIGHLIGHT

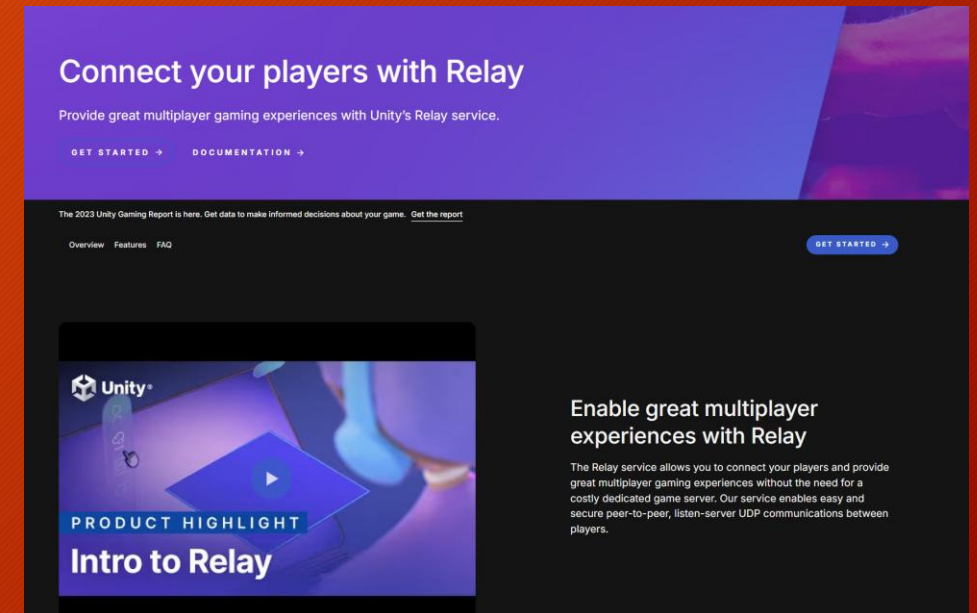
Intro to Lobby

Connect players for multiplayer fun

Lobby allows your players to connect before or during a game session. Players create public lobbies using simple game attributes which other players can then search, discover, and join. Invite-only lobbies also allow players to create private spaces for select participants only.

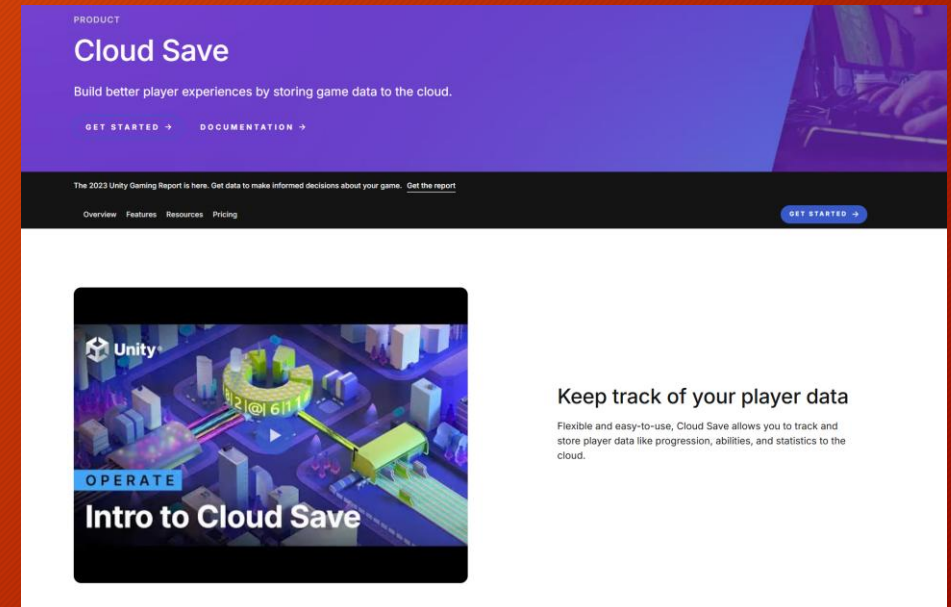
Connecting players with Relay

- <https://unity.com/products/relay>
- Acts as a proxy to connect players
- Based on UDP packets
- Max of 100 concurrent players in the same session
- Based on Join Codes



Persistent Data with Cloud Save

- <https://unity.com/products/cloud-save>
- Makes player's data independent of device
- Players can access data across devices
- Mitigate data loss when player change device



Event Driven Programming

- Events are typically used to signal user actions.
- The publisher determines when an event is raised.
The subscribers determine what action is taken in response to the event.
- An event can have multiple subscribers.
A subscriber can handle multiple events from multiple publishers.
- Events that have no subscribers are never raised.
- When an event has multiple subscribers, the event handlers are invoked synchronously when an event is raised.

Useful resources

- House's blog post "Choosing the right netcode for your game" (<https://blog.unity.com/games/how-to-choose-the-right-netcode-for-your-game>) reviews different multiplayer solutions
- Unity Gaming Services Use Cases (<https://github.com/Unity-Technologies/com.unity.services.samples.use-cases>)
- Boss Room (<https://docs-multiplayer.unity3d.com/netcode/current/learn/bossroom/bossroom/>), fully functional co-op multiplayer RPG made with Unity Netcode
- Unity Services ([Unity Gaming Services: Live Game Development Platform](#))